Houston Community College

AI Agent:

FitForm

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ITAI 2376 – Deep Learning in Artificial Intelligence

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FitForm Agent

Problem Statement

Despite the widespread use of fitness trackers and workout apps, many users still struggle to find systems tailored to their specific body type, lifestyle, and metabolic profile. Research from the National Academy of Sports Medicine (NASM) highlights that understanding one's somatotype can reveal unique structural and physiological challenges that come with each body type (Payne, 2025). While existing platforms track workouts, nutrition, and movement, they lack contextualized feedback that adapts to the user's somatotype. To bridge this gap, I propose an AI-powered agent designed to guide individuals across diverse body types through personalized form insights, adaptive goal-setting, and real-time support to help them overcome somatotype-specific barriers and achieve sustainable fitness outcomes.

Project Option

The fit form agent will be an interactive learning companion. My agent acts like a fitness coach that learns about the user's body type and guides them on a personalized wellness journey, adjusting recommendations and eventually giving smart feedback.

Agent Design & Tool Selection

The goal of this agent is to combine a Deep Q Network with a structured profiling for custom workouts, dietary, and lifestyle recommendations.

DQN Agent Components

Component	Role	Tools/Frameworks
Environment	The state space (user body type, user posture, reps, fatigue levels, metabolic health, etc) and actions (guidance, corrections, advice)	Custom Gym-like wrapper
User Profile	Captures body type, estimated metabolism and BMR, fitness goals, and dietary restrictions.	Python
State Representation	Converts user data into structured tensors	OpenCV, NumPy
Replay Buffer	Stores past transitions for training stability	Replay Memory class
Q-Network	Neural network estimating Q-values for each action	Tensorflow

Target Network	Stabilizes learning by slowly syncing with Q-Network	Periodic weight update strategy
Epsilon-Greedy Policy	Balances exploration and exploitation	Decay schedule (e.g. exponential decay)
Training Loop	Run episodes, updates Q-Network, computes loss	Optimizer step
Rewards System	Measures success of agent's actions (e.g. improved form = positive reward)	Custom reward logic
Action Mapping	Maps outputs to user-visible recommendations (e.g.)	NLP or UI prompt generator
Dietary Guidance Module	Generates nutrition suggestions based on body type, goals, meal timing, micro, macros, hydration	Rule-based NLP
Lifestyle Recommender	Suggests wellness habits	Calendar and alert integration

FitForm AI integrates reinforcement learning with user-specific profiling, evolving alongside the user to promote holistic well-being.

Development Plan

Week 1: Core Development & Foundation

Day 1-2:

- Finalize user profiling logic
- Draft clear input forms and mock user data samples
- Outline reward structure (workout, nutrition, consistency)

Day 3-4:

- Implement pose analysis + state tensor formation
- Set up DQN framework
- Define action space

Day 5-6:

- Build dietary guidance module with body type logic
- Structure early rule-based suggestions
- Test basic Q-network flow with mock environment

Week 2: Testing, UI, & Final Polish

Day 8–9:

- Add lifestyle recommender hooks (sleep, stress, habits)
- Refine action mapping for clarity and tone

Day 10–11:

- Test full reward feedback loop
- Tune decision-making (adjust Q-learning parameters)
- Integrate logging and transparency (keyword summaries, model insight)

Day 12–13:

- Draft and refine documentation (README, setup, usage tips)
- Add visual aids (flowchart, architecture sketch)
- Optional: Compare deployment frameworks (Gradio)

Day 14:

- Final debug pass and polish
- Run final demo walkthrough or record a short video

Evaluation Strategy

1. Functional Success:
   a. Agent Accuracy
   b. Environment Responsiveness
   c. Profile Adaptivity
2. Learning Performance
   a. Exploration Balance
   b. Reward Curve
3. Project Delivery

Resource Requirements

1. Github Pages
2. Google Colab for development and GPU usage during training
3. Hugging Face

Risk Assessment

Challenges that I anticipate and the plan to address them include:

1. **Coding challenges**: I will use a coding assistant and start early so I have time for debugging.
2. **Defining meaningful rewards**: I have to translate “feeling better” “looking better” “consistency” to numbers. This part will be tricky.

3. **Balancing Personalization vs. Generalization**: Every person is different. I want to designing an algorithm that respects uniqueness that doesn't become overfitted or won't work all together.

References

Agent Planning & Orchestration. (2025). *ITAI 2376 Mod 12*. HCC.

Payne, A. (2025). *Body Types: Mesomorph, Ectomorphs, and Endomorphs Explained*. Retrieved from NASM: https://www.nasm.org/resource-center/blog/body-types-how-to-train-diet-for-your-body-type?srsltid=AfmBOorwHaToiEWKJFq6xrLj3h6ssBpzibZp3sKF6NZLS7xKESfkLQEN#how-to-identify-body-type

Reinforcement Learning & AI Agents. (2025). *ITAI 2376 Module 10*. HCC.